

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I N. Chawran Aryan (192365053) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

N. Chawran Aryan

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1: Smart Home Automation System using Intelligent Agents

Problem Statement

A smart home automation system is designed to control lighting, temperature, and security based on user behavior and environmental conditions. The system receives real-time inputs such as motion detection, temperature changes, and time of day. It should also learn user preferences over time to improve comfort, energy efficiency, and security.

Objective

1. To understand the different types of intelligent agents.
2. To implement a smart home automation system using AI concepts.
3. To analyze the working of Simple Reflex, Model-Based, Goal-Based, and Utility-Based agents.
4. To improve user comfort, energy efficiency, and home security.
5. To identify the most suitable intelligent agent for the system.

Solution

Step 1: Identify the Inputs

The smart home receives the following inputs:

- Motion detected
- Temperature
- Time of day
- User preferences

Example:

Motion = Yes

Temperature = 32°C

Time = Night

Step 2: Analyze Current Situation

The system checks the sensor values.

- Motion is detected.
- It is night.
- Temperature is high.

Step 3: Apply Simple Reflex Agent

The Simple Reflex Agent follows predefined rules.

Rules:

- If motion is detected at night → Turn ON lights.
- If temperature > 30°C → Turn ON AC.
- If no motion at night → Activate security alarm.

Result:

Lights ON

AC ON

Security Monitoring

Step 4: Apply Model-Based Agent

The Model-Based Agent remembers previous information.

Example:

Yesterday

User entered room at 8 PM.

Today

At 8 PM motion detected again.

The agent predicts the user's routine and turns on the lights automatically.

Step 5: Apply Goal-Based Agent

Goal:

Provide comfort while reducing electricity usage.

Actions:

- Turn lights ON only when required.
- Maintain comfortable room temperature.
- Lock doors automatically.

Step 6: Apply Utility-Based Agent

The Utility-Based Agent compares different actions.

Example:

Option 1

Keep AC ON all night

Comfort = High

Electricity = High

Option 2

Set AC to 24°C

Comfort = High

Electricity = Low

The agent selects **Option 2** because it provides maximum overall benefit.

Step 7: Final Decision

The system performs the following actions:

Lights ON

AC ON

Security Monitoring

Save user preference

Step 8: Final Result

The home becomes

- Comfortable
- Energy Efficient
- Secure

Python Code

```
motion = True
```

```
temperature = 32
```

```
time = "Night"
```

```
print("Smart Home Automation System")
```

```
if motion and time == "Night":
    print("Lights ON")
else:
    print("Lights OFF")

if temperature > 30:
    print("Air Conditioner ON")
elif temperature < 20:
    print("Heater ON")
else:
    print("Temperature is Comfortable")

if not motion and time == "Night":
    print("Security System Activated")
else:
    print("Security Monitoring")
```

Output

Smart Home Automation System

Lights ON

Air Conditioner ON

Security Monitoring

Discussion

- The system uses sensor data to make intelligent decisions.
- A Simple Reflex Agent reacts only to current inputs.
- A Model-Based Agent remembers previous states.
- A Goal-Based Agent works toward achieving user comfort and safety.
- A Utility-Based Agent selects the action that provides the highest overall benefit.
- A Utility-Based or Hybrid Agent is most effective because it balances comfort, energy efficiency, and security.

Conclusion

The Smart Home Automation System successfully demonstrates the application of intelligent agents. Among all agent types, the Utility-Based Agent provides the best performance by making balanced decisions based on multiple factors, resulting in improved comfort, energy efficiency, and security.

Question 2: Robot Navigation in an Unknown Maze using UCS and IDS

Problem Statement

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the environment. The robot should use uninformed search strategies such as Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) to explore the maze and reach the exit efficiently.

Objective

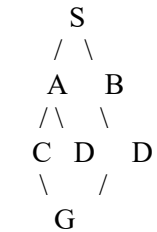
1. To understand uninformed search strategies.
2. To implement Uniform Cost Search (UCS).

3. To understand the working of Iterative Deepening Search (IDS).
4. To compare UCS and IDS based on time and space complexity.
5. To identify the best search strategy and intelligent agent for maze navigation.

Solution

Step-by-Step Solution

Assume the following graph:



Edge Costs

$S \rightarrow A = 1$

$S \rightarrow B = 4$

$A \rightarrow C = 2$

$A \rightarrow D = 5$

$B \rightarrow D = 1$

$C \rightarrow G = 3$

$D \rightarrow G = 2$

Uniform Cost Search (UCS)

Step 1

Start from S.

Priority Queue

$[(S, 0)]$

Visited

None

Step 2

Remove node with minimum cost.

Expand

S

Visited

S

Insert neighbours.

$A = 1$

$B = 4$

Queue

$[(A, 1), (B, 4)]$

Step 3

Expand A.

Visited

S,A

Insert

C = 3

D = 6

Queue

[(C,3),(B,4),(D,6)]

Step 4

Expand C.

Visited

S,A,C

Insert

G = 6

Queue

[(B,4),(D,6),(G,6)]

Step 5

Expand B.

Visited

S,A,C,B

New path

B → D

Cost = 5

Better than

D = 6

Update Queue

[(D,5),(G,6)]

Step 6

Expand D.

Visited

S,A,C,B,D

New Path

D → G

Cost = 7

Existing

G = 6

Ignore new path.

Queue

[(G,6)]

Step 7

Expand G.

Goal Found.

Optimal Path

S → A → C → G

Total Cost

1 + 2 + 3 = 6

Iterative Deepening Search (IDS)

Depth = 0

Visit

S

Goal?

No

Depth = 1

S

└─ A

└─ B

Goal?

No

Depth = 2

S

└─ A

| └─ C

| └─ D

└─ B

└─ D

Goal?

No

Depth = 3

S

└─ A

| └─ C

| | └─ G ✓

| └─ D

| └─ G

└─ B

└─ D

└─ G

Goal Found.

Final Path

$S \rightarrow A \rightarrow C \rightarrow G$

Python Code (Uniform Cost Search)

```
import heapq

graph = {
    'S': [('A',1), ('B',4)],
    'A': [('C',2), ('D',5)],
    'B': [('D',1)],
    'C': [('G',3)],
    'D': [('G',2)],
    'G': []
}

def ucs(start, goal):
    queue = [(0, start, [])]
    visited = set()

    while queue:
        cost, node, path = heapq.heappop(queue)

        if node in visited:
            continue

        visited.add(node)
        path = path + [node]

        if node == goal:
            print("Optimal Path:", " -> ".join(path))
            print("Total Cost:", cost)
            return

        for neighbor, weight in graph[node]:
            if neighbor not in visited:
                heapq.heappush(queue, (cost + weight, neighbor, path))

ucs('S', 'G')
```

Output

Optimal Path:

S -> A -> C -> G

Total Cost: 6

Discussion

- Uniform Cost Search expands the node with the lowest path cost.
- Iterative Deepening Search combines the advantages of DFS and BFS while using less memory.
- UCS guarantees the optimal solution when path costs differ.
- IDS is memory efficient but may revisit nodes multiple times.

- A Goal-Based Agent using UCS is the most effective combination for finding the optimal path in an unknown maze.

Conclusion

The robot successfully finds the optimal path using Uniform Cost Search. UCS guarantees the minimum-cost solution, while IDS offers memory efficiency for large search spaces. Combining a Goal-Based Agent with UCS provides intelligent and optimal navigation in unknown environments.